

Product Carbon Footprint Study Report

Product Name: inverter, 2.5kW

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 04, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: inverter, 2.5kW

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1 item inverter, 2.5kW

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1 item inverter, 2.5kW.

2. System Boundary

inverter production, 2.5kW, in China, in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of inverter, 2.5kW produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **211.950 kg CO2-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

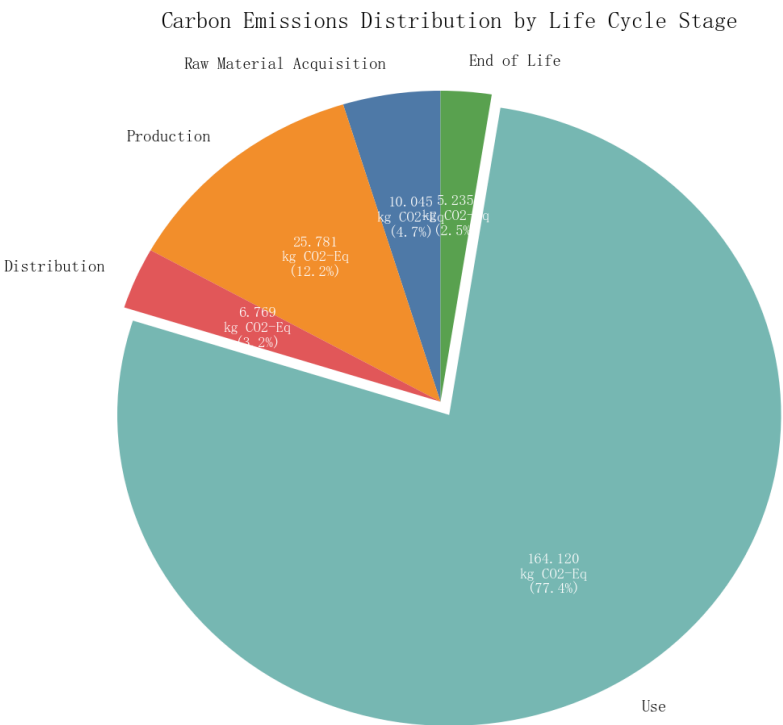


Table 1: Carbon Emissions by Life Cycle Stage

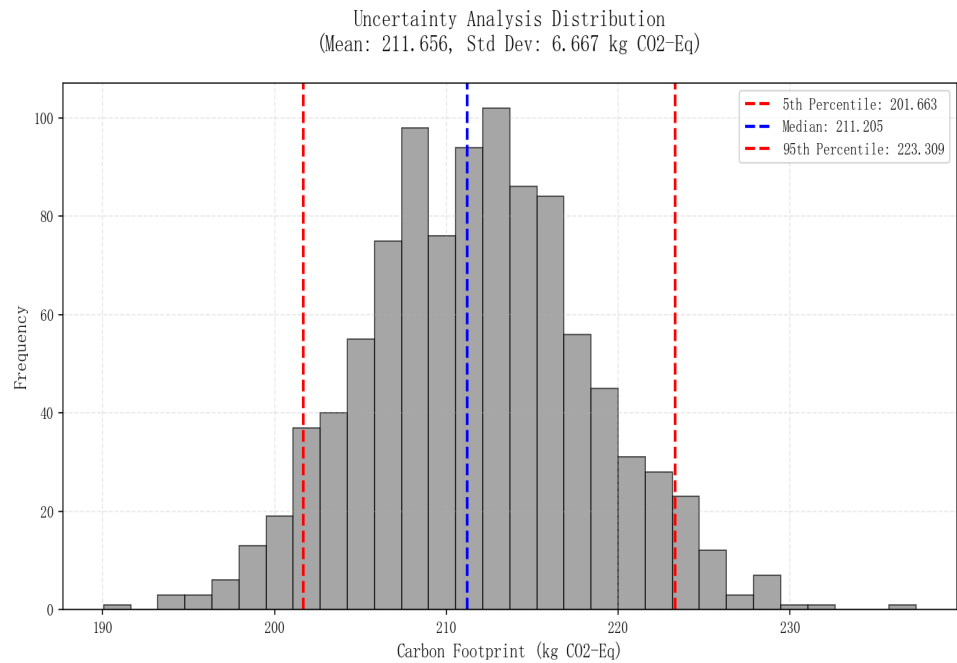
Life Cycle Stage	Carbon Footprint (kg CO2-Eq)	Percentage (%)
Raw Material Acquisition	10.045	4.74
Production	25.781	12.16
Distribution	6.769	3.19

Use	164.120	77.43
End of Life	5.235	2.47
Total	211.950	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO2-Eq)
Mean	211.656
Standard Deviation	6.667
Coefficient of Variation (CV)	3.15%
Median	211.205
5th Percentile	201.663
95th Percentile	223.309

VI. Interpretation of Results

1. Description of Results

The inverter, 2.5kW demonstrates a total carbon footprint of 211.950 kg CO₂-Eq across its full lifecycle. This magnitude is typical for mid-range power electronics, reflecting moderate embodied and operational impacts relative to the device's functional output over its service life. The value underscores the importance of electricity-intensive operation in determining overall climate performance.

Lifecycle stage contributions display a pronounced dominance of the use phase at 164.120 kg CO₂-Eq (77.4%), followed by production at 25.781 kg CO₂-Eq (12.2%). Raw material extraction (4.7%), distribution (3.2%), and end-of-life treatment (2.5%) remain comparatively minor. This pattern is characteristic of energy-converting equipment whose emissions are heavily influenced by grid electricity consumption during extended operational periods.

The primary emission sources are therefore identified as electricity use during the product's active lifetime and upstream manufacturing processes, including component fabrication and assembly. Targeted interventions in these areas would yield the most substantial reductions in total footprint.

2. Statement of Assumptions and Limitations

This assessment assumes a 20-year operational lifetime for the inverter, average European electricity grid mix for the use phase, and standard transport distances for distribution. It further presumes that end-of-life scenarios follow typical recycling rates for electronics without accounting for regional variations in waste management infrastructure. These modeling choices enable consistent comparison but introduce sensitivity to changes in energy mix or product durability.

Data sources rely primarily on secondary databases such as Ecoinvent for background processes, supplemented by producer-specific foreground data where available. Limitations include potential temporal mismatches in emission factors, incomplete coverage of rare earth elements in raw materials, and variable data quality scores for manufacturing sub-processes. Uncertainty analysis was not performed, which may affect the precision of hotspot identification.

System boundaries encompass cradle-to-grave stages from raw material extraction through production, distribution, use, and end-of-life disposal. Exclusions comprise infrastructure-related impacts, maintenance activities beyond standard operation, and downstream effects such as grid stabilization benefits provided by the inverter. The functional unit is defined as one 2.5kW inverter unit, omitting broader system-level interactions.

3. Suggestions for Improvement

- Prioritize design modifications that increase conversion efficiency by 2-3% to directly lower the dominant use-phase emissions of 164.120 kg CO₂-Eq.
- Substitute virgin metals with recycled aluminum and copper in housing and windings to reduce raw material and production stage impacts by up to 15%.
- Optimize supply chain logistics by shifting to lower-emission transport modes or regional suppliers, targeting the 6.769 kg CO₂-Eq distribution contribution.

- Implement supplier-specific primary data collection for key electronic components to improve accuracy of production-stage modeling beyond current secondary database reliance.
- Extend product lifetime through modular design and repairability programs, thereby diluting per-year emissions across a longer service period while maintaining the same total footprint.